Worksheet 18

November 4, 2026 Name: _____

About 10 minutes total. Begin independently, then discuss. Show your work.

1. A triangle and a quadrant

Use the principal range to identify the quadrant. Then find the requested exact value.

(a) Let $\alpha = \arctan\frac{5}{12}$. Identify the quadrant of α and find $\sin\alpha$.

(b) Let $\beta = \arccos\left(-\frac{7}{25}\right)$. Identify the quadrant of β and find $\tan\beta$.

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2. Other inverse trigonometric functions

Find each exact principal value in radians. Use arccot with range $(0, \pi)$.

(a) $\operatorname{arcsec}(-\sqrt{2})$

(b) $\operatorname{arccsc}(2)$

(c) $\operatorname{arccot}(-\sqrt{3})$